Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

Xinya Du, Assistant Professor, Computer Science, The University of Texas at Dallas xinya.du@utdallas.edu https://xinyadu.github.io

Program Officer: Dr. Behzad Kamgar-Parsi, Code 311 — Artificial Intelligence / Machine Learning, Reasoning and Intelligence

Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

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Problem. A tool-using AI agent can follow a plausible plan while making a single consequential mistake: it misreads an image, selects a wrong entity in a database, or passes a time in the wrong time zone to a navigation calculation. Outcome-based training reveals only that the final plan failed. Generic self-critique often cannot pinpoint the first unsupported step or tell what evidence would resolve it. Consider a port-entry assistant that reads the right tide table but uses local time as UTC; the error is neither a missing fact nor a defective calculator, but a violated precondition.

Central hypothesis. Agents trained to identify the type, location, and evidence behind their own mistakes will make better next-action decisions than agents trained only on final outcomes or an undifferentiated process score. The new object is learned self-assessment across tool calls and multimodal observations, rather than static claim checking or scientific-rule applicability.

Research plan.

- **Build checkable records of agent actions.** Represent each step as an action or claim paired with its evidence, preconditions, and observed result. Text passages, structured knowledge, tool outputs, and image regions will have distinct evidence types. Synthetic multistep examples from entity and event relations will provide controlled training cases, while unlabeled trajectories will expose diverse natural behaviors.
- **Learn typed self-assessment from synthetic errors.** Inject controlled fabrication, entity mismatch, invalid inference, wrong-tool-argument, visual-attribute, and relational errors into otherwise plausible trajectories. Train a verifier to classify the failure type and localize the earliest consequential error. We will test transfer from synthetic to natural errors and calibration when a new visual scene or tool is encountered, adapting the fine-grained feedback of our vision-language work.
- **Turn self-assessment into selective action.** Use the agent's diagnosis to decide whether to re-read evidence, inspect an image crop, fix a tool input, backtrack, or ask for help. Process rewards can guide trajectory improvement, but we will penalize unnecessary retries and missed deadlines. This makes self-assessment a component of planning, not merely an explanation produced after the answer.

Evaluation and expected outcomes. MEQA and MuSiQue will test text/knowledge composition, while ScienceQA and a tool-use testbed with controlled argument errors will test multimodal grounding and corrective actions. We will measure typed-error localization, calibrated self-assessment, repair success, and task success per tool call against outcome-only training and untyped PRMs.

Naval relevance. A maritime support agent combining chart imagery, arrival notices, and a tide calculation must be able to notice a wrong image reference or time conversion before recommending an action. Typed self-assessment can trigger the particular recheck required instead of either trusting a bad plan or restarting everything. The work matches this portfolio's explicit interests in reasoning under uncertainty, self-assessment, visual understanding, and integration of knowledge with planning.

Prior work and PI qualifications. Our FG-PRM (Li et al., 2025) and HiPRAG (Wu et al., 2026) develop process-level feedback for reasoning and retrieval; FaithScore (Jing et al., 2024) and FGAIF (Jing and Du, 2025) address visual-language grounding. The PI holds a 2024 NSF CAREER award and led integration of the RESIN team's high-performing DARPA KAIROS delivery system as a postdoc.